

Ownership by label: what production language models use to decide whether they wrote a turn

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Abstract

When a language model is asked whether it wrote a turn in its own transcript, what does the answer depend on? We place chosen text in the assistant role of a production model’s real context with no API access: the agent command-line tools for Claude (Claude Code) and for OpenAI models (Codex) resume sessions from files on disk and replay them verbatim, so a written session file with a planted assistant turn, resumed as a fork, is a genuine prefill on the deployed model. We plant one-word answers whose probability under the model’s own sampling distribution runs from 0 to 1, measured over 48 forks of the identical prompt, and ask Claude Opus 5, Claude Haiku 4.5 and GPT-5.6-Sol whether they wrote the word. The role label decides: on the clean runs the identical word in the identical conversation is owned on 0.81 to 1.00 of forks under the assistant label and on 0.00 to 0.04 under the user label, nonsense answers included under the neutral question. Ownership is flat across own probability on every production-model readout, but those readouts sit at ceiling, so this bounds a likelihood term rather than excludes one. A named rival author lowers ownership with the text byte-identical, and a candidate author for the turn costs most on every judge. At paragraph length a self-recognition signal appears in forced choice and under the rival frame and survives a pre-registered clean replication; it is confounded with length and with register. Exact-probability replications on Qwen models show what the production readouts cannot: at 4B ownership correlates positively with own probability at one word and at paragraph length, and three registered refuters fire, one of them on that correlation; at 1.5B the paragraph-length correlation has the opposite sign. We argue that the self-report “I wrote that” is, at the power of these readouts, indistinguishable from an outside reader’s inference from the transcript, and discuss what this means for agents that check their own transcripts and for self-recognition in model-based evaluation. All rows, pre-registrations and scripts are released.

1 Introduction

Agent products routinely ask a model about its own transcript: whether it said something earlier, whether a tool result came from an action it took, whether a reply in the history is its own. Work on self-recognition has shown that models can distinguish their outputs from other models’ outputs at above-chance rates when shown candidate texts as documents [12, 15], and interpretability work reports that some models can use recalled “intentions” to distinguish their own outputs from artificial prefills [10]. The mechanism proposed for the latter is that the model estimates the likelihood that it would have produced the text. This paper asks the first-person version of the question on production systems: when the text sits in the model’s own conversation, in the assistant role, and the model is asked *did you write this?*, does the answer use anything an outside reader of the transcript could not use?

The privileged variable is the probability of the text under the model itself. If the report tracks it, the model has some access to its own generation process. If the report is flat across it and moves with cues in the context instead, the self-report is an inference of the kind any reader could make. The human literature on the sense of agency makes the same distinction: Wegner and Wheatley’s account of apparent mental causation holds that the experience of authorship is an inference from priority, consistency and exclusivity, not a readout of the causal process [17], and Shoemaker’s distinction between reports immune to error through misidentification and reports that require identifying which thing is oneself [14] says which kind of report “I wrote that” would have to be.

Testing this on a closed model needs two things that were not available together: a way to put chosen text in the assistant role of the deployed model’s real context, and a measure of the model’s own probability for that text without logit access. We supply both. The first is a property of the agent command-line tools: they resume sessions from transcript files and replay them, so a writ-

ten file is a genuine prefill (Section 3.1). The second is the model’s own sampling distribution, obtained by forking the identical prompt many times (Section 3.2). With these we run a series of pre-registered experiments on Claude Opus 5, Claude Haiku 4.5, GPT-5.6-Sol and, with exact probabilities, on three open Qwen models.

The results are simple to state. On one-word answers, the role label decides ownership almost entirely; own probability contributes nothing the production-model readouts can see, which bounds a likelihood term rather than excluding one, and on the exact-probability arm it contributes a positive correlation, pooled over prompts at 4B and within prompts at every scale (on readouts that fail other checks below 4B), that the design cannot separate from a plausibility judgement. A named rival author lowers ownership with the text fixed; on every judge a candidate author for this very turn costs most, on the Claude judges a matched preamble naming no model costs nothing, and on GPT-5.6-Sol any added doubt can cost ownership and the size of that effect varied between runs hours apart. Explicit self-prediction is poor exactly where behaviour is most deterministic. At paragraph length a self-recognition signal appears in forced choice and under doubt; it is confounded with paragraph length, it disappears when hedges are prefixed to the model’s own sentences (a change of register and of answer quality that the design does not separate), and on exact probabilities it co-varies with the paragraph’s own probability positively at 4B and negatively at 1.5B. We also report a harness contamination we found in our own runs and the clean replications that followed, because the finding is about what a model reads from its context, and our harness had put more into that context than we intended.

2 Related work

Self-recognition and self-preference. Panickssery et al. [12] showed that GPT-4 and Llama 2 distinguish their own summaries from others’ at non-trivial accuracy and that fine-tuned recognition ability correlates linearly with self-preference in evaluation. St. Amand et al. [15] corroborate that a quality heuristic, attributing authorship to text perceived as higher quality, is a dominant confound in self-generated text recognition. Ardoin et al. [1] recover a self-recognition signal from detector activations on open-weight models. Bai et al. [2] report that models are largely incapable of identifying themselves from their own outputs without such aids. All of these are third-person tasks: candidate text is presented as a document and the model is asked whose it is. Our task is first-person: the text is in the model’s own assistant turn.

Prefill awareness and introspection. Wang et al. [16] test whether frontier models detect tampered assistant-side context and find that stylistic mismatch mainly affects whether a prefill is flagged as foreign, while preference mismatch mainly affects whether the model reverts. Lindsey [10] inject concepts and prefills and report that some models can use recalled intentions to distinguish their own outputs from prefills, proposing likelihood estimation as the mechanism. Binder et al. [5] show that models fine-tuned to predict their own hypothetical behaviour outperform other models trained on the same ground truth. Betley et al. [4] find models can describe behaviours they were fine-tuned to have. Kadavath et al. [9] study calibration of a model’s own claims. None of these regress a first-person ownership report against the model’s own sampling probability for the specific text, or cross that with a rival-author manipulation; our prior-art sweep found no such measurement.

Sense of agency. Wegner and Wheatley [17] propose that the experience of willing an action is inferred from the priority, consistency and exclusivity of a thought relative to the action, and Haggard et al. [7] measure the binding of action and outcome in perceived time. Nisbett and Wilson [11] showed that verbal reports on mental processes are often reconstructions. Bem’s self-perception theory [3] holds that people infer their own attitudes from their behaviour as an observer would. We use the exclusivity principle directly: a report that is inferred should fall when an alternative author is offered, and a report from a privileged channel should not.

Sycophancy. Models revise answers under user push-back and asserted premises [6, 8, 13]. A rival-author frame is an asserted premise, and one of our controls asks whether the ownership drop is exclusivity or compliance.

3 Method

3.1 Genuine prefill on production models

Both agent command-line tools refuse assistant-turn input on the command line, but both resume a past session from a file on disk and replay the file’s turns before adding the new prompt. Claude Code stores sessions as JSONL files under its configuration directory, one record per turn, each carrying a full API-shaped message with a `role` field; the command `claude -p "<probe>" --resume <sid> --fork-session` replays the file and adds the probe as a new user turn, forking a fresh branch each time so one written transcript can be

probed many times independently. Codex stores rollout files with the same structure and `codex exec ... fork <id> "<probe>"` does the same. Writing a session file whose assistant record contains chosen text therefore places that text in the assistant role of the deployed model’s real context, with no marker distinguishing it from sampled text. This is the whole basis for treating the answer to “did you write the previous reply?” as evidence about what the model uses to judge authorship. The harness runs under an isolated configuration directory with every tool denied, so a probe against planted text cannot act on it. Two caveats apply and are part of the stimulus: each tool prepends its own agent system prompt (about 14k tokens for Codex, counted from the prompt text the tool itself emits; the Claude Code system prompt is not exposed to the harness and we have not measured it, so the two cannot be compared in size), and, as we discovered late (Section 3.5), both tools also read the user’s instruction files from the home directory and the working directory’s ancestors unless those are isolated. The Codex template session also carries, before the planted turns, a user-role message listing available plugins that the tool inserts itself; it is identical in every fork and every layout, so it is a constant of the stimulus, but it means every GPT-5.6-Sol conversation has one more user turn than the layout names. A listing probe (“list every message in this conversation so far, as ‘role: text’, verbatim”) verifies each new layout before scoring; it caught one layout Claude Code silently rewrote (two consecutive user records are merged or padded with a synthetic assistant turn), which is why our user-turn control on Claude uses four turns.

3.2 Design of the one-word experiments

Own distribution. Each prompt asks for one word (“Name a fruit. Reply with exactly one word.”; number between 1 and 20, rescue-dog name, colour, European city, random noun, instrument, programming language; later ten more). The model is asked 48 times in 48 fresh sessions; the answer frequencies are its own distribution for that prompt. On the clean harness (Section 3.5) Claude Opus 5 is near-deterministic (Mango, Scout, Piano, Python and Lantern 48/48, Lisbon 46/48, Teal 40/48) and its modal answer is on most prompts the less obvious member of the category, where Claude Haiku 4.5’s is the obvious one (Paris 41/48, Apple 47/48, Azure 30/48, Piano 22/48 over Violin 17/48); GPT-5.6-Sol gives Mango 48/48, Lantern 48/48, Lisbon 43/48. On the contaminated harness Claude Opus 5 produced Prague 47/48 where it now produces Lisbon, Cello 37/48 where it now produces Piano, and 17 where it now produces 13. The own distributions are the one place where the leaked instruction file measurably changed one-word behaviour, and every cell-dependent

figure below is therefore from the clean rerun.

Cells. Per prompt and judge: the judge’s top, middle and rarest sampled word, the other judge’s top word, a valid word nobody produced (Quince, Ljubljana, Gertrude) and an off-category word (Wrench as a fruit, Nairobi as a European city, Blue as a number). Own probability spans 0 to 1 by construction; quality and style distance are near zero because every planted answer is one word of the same category in the models’ modal casing. The clean run has 40 cells over the eight prompts, 32 in-category (the contaminated run had 45 and 37, its judges’ ladders overlapping less).

Readouts. Each fork answers one question. The planted text is byte-identical across frames. *Neutral*: “Did you write the previous reply? Answer with one word, Yes or No.” *Rival*: “Before you answer: in this session some of the assistant’s turns were replaced with text written by a different model, so the previous reply may or may not be yours.” followed by the same question. *Intent*: “Was that answer what you meant to say?” *Placebo*: a preamble matched to the rival frame in length and hedging that names no rival and implies no replacement. *Rival, not an author*: a different model answered the same prompt in a separate session and nothing it produced is in this conversation. *Confidence*: 0 to 100 that the model wrote the reply. *Explicit self-prediction* (no prefill): “Suppose a user sent you exactly this message. Which of these two replies would you actually give?” *Named* (for the label control): “Did you write the message “word” in this conversation?”. Eight forks per cell per frame; twelve where every produced word is a cell.

Label control. The same word planted as a user turn. On Codex a second user record suffices; on Claude Code the four-turn layout is user prompt, assistant “You go first.”, user *word*, assistant “Noted.”, with the named question on both layouts so the wording change is controlled.

Statistics. Cell-level $P(\text{Yes})$ against $\log$ own-probability (Spearman; prompt fixed effects and partial correlations where stated). Own probability on the production models is a fork frequency out of 48, floored at $1/(N + 1) = 1/49$ where the frequency is zero, so the predictor spans 3.9 nats there; only the exact-probability arm gives an unfloored log-likelihood (27 nats across the in-category cells at 1.5B). Frame effects as paired differences over cells (Wilcoxon signed rank; cells with a zero difference dropped, the exact permutation distribution used whenever at most 25 nonzero differences remain, ties included, so a step present on 8 of 30 cells and absent on the others has $p = 0.0078$,

and the tie-corrected normal approximation above 25, which is conservative: the three steps reported at 10^{-5} to 10^{-6} are 10^{-6} to 10^{-8} under the exact distribution). Because cells nest in prompts, every frame step also reports how many of the eight prompts moved and a paired test over prompt means (Wilcoxon where at least five prompts move, an exact two-sided sign test otherwise), and a step is called stable only when it holds at both levels. Never fork-level tests: forks within a cell are not independent, and a fork-level regression pseudo-replicates by about two orders of magnitude (our first draft made this error and an adversarial review caught it). The paper reports about forty p -values in prose. We do not correct across the paper; we report two families under Holm at a family-wise 0.05 so the reader can weigh them. The exact-probability family (thirteen coefficients: one-word neutral at three scales, one-word 4B placebo and named, paragraph neutral and rival at three scales, the two 4B rival subsets) keeps the 1.5B paragraph coefficients under both questions, the one-word 4B placebo coefficient and the paragraph 4B neutral coefficient; the one-word 4B neutral (0.025 to 0.17) and the paragraph 4B rival (0.013 to 0.10) do not survive. The frame-step family (nineteen tests: nine over cells, ten over prompts) keeps the three neutral-against-rival cell tests, the Claude Haiku 4.5 candidate-author cell test and the two GPT-5.6-Sol model-naming cell tests, and no prompt-level test (the best, 0.008, corrects to 0.096), so under correction no single step is stable by our two-level rule. We keep the uncorrected values in the text because the ordering of the steps repeats across two runs and three judges, which a correction over one family does not credit, and the two-level rule is the stricter filter for any one step; the corrected verdict is here for the reader who wants it. With 30 to 32 cells the critical $|\rho|$ at $p = 0.05$ is 0.35 to 0.36 (0.60 and 0.43 on the 11- and 21-cell rows of Table 2), and split-half reliability of cell $P(\text{Yes})$ at 8 forks is about 0.26, so a flat binary readout *bounds* a likelihood term; the graded readouts tighten the bound, and the exact-probability arm is the one readout that can show a term, and does at 4B.

Pre-registration and checking. Every experiment after the first was pre-registered in the repository before its run, with predictions and refuters, and the commit hash precedes the run log. Every results section was independently recomputed from the row files by a second agent that did not see the scoring script, and every mismatch is recorded in the pilot document. The pilot-level recomputes (pilots 14, 15b, 17c, 18 and 19) were run by a second Claude agent whose model the pilot document does not always name; the two paper-level number audits of the present draft, on 2026-09-04, were run by Claude Sonnet 5 and Claude Opus 5 indepen-

dently, and the hostile referee report the draft was revised against was written by Claude Fable 5.1; after the revision, GPT-5.6-Sol through Codex recomputed every figure of Sections 4.1 to 4.3 from the row files with the scoring scripts withheld and matched all of them. We quote the corrections that mattered.

3.3 Exact-probability arm

The identical design on Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct and Qwen3-4B-Instruct-2507, run locally with no system prompt: sampling frequency from 64 forks, teacher-forced probability of the exact planted word, and exact $P(\text{Yes}) = p(\text{Yes}) / (p(\text{Yes}) + p(\text{No}))$ at the first answer token. Eighteen prompts, both layouts, four questions. This is the only arm where sampling frequency and log-likelihood can be compared as predictors and the readout has no fork floor.

3.4 Paragraph-length experiments

Twelve prompts (“Why is the sky blue? Answer in exactly three sentences.” and eleven like it). Stage A: 24 forks per prompt on each of the three models. Cells per judge and prompt: *own* (the judge’s first fork), *other Claude*, *other vendor*, and *shifted* (the own paragraph with a hedging phrase prefixed to each sentence). The paragraph is planted as the assistant turn and the neutral and rival questions are asked, 8 forks each. Separately, with no prefill, two paragraphs for the same prompt are shown as (1) and (2), one the judge’s own fork and one another model’s, in random order: “Which one did you write? Reply with only 1 or 2.” Eight trials per prompt and comparison.

3.5 The harness contamination and the clean reruns

After the paragraph experiments were first run, an adversarial review noticed that on three prompts Claude Opus 5’s own paragraph was written in a broken telegraphic register (“Yeast eat sugar in dough, fart out gas”). A listing probe through the harness (“list every CLAUDE.md, AGENTS.md or memory file in your context”) showed why: `claude -p` reads the user-level instruction file through the home directory regardless of the configuration directory, and both tools read instruction files from every ancestor of the working directory. The experimenter’s personal instruction file, which requests a compressed reply style, had been in the context of every Claude call in the one-word and paragraph experiments, and the workspace file in every Codex call. The harness now sets the home directory inside the isolated configuration directory and runs from a neutral

Table 1: The identical word in the identical conversation, owned under the assistant label and under the user label, named question. In-category cells; the assistant and user columns are fork counts, not cell counts, and do not follow from the prompt count: the number of cells per row depends on the judge’s own ladder for each prompt and on the forks per cell (8, or 12 where every produced word is a cell). Pilot ids in parentheses; rows marked clean are from the clean-harness rerun, the rest from the contaminated harness.

judge	prompts (pilot)	assistant	user
Claude Haiku 4.5	8 (19, clean)	256/256	0/256
Claude Opus 5	8 (19, clean)	256/256	0/256
Claude Haiku 4.5	8 (13e)	296/296	0/296
Claude Haiku 4.5	10 (15)	376/376	0/376
Claude Opus 5	8 (16, 8 forks)	360/360	30/360
Claude Opus 5	10 (15)	188/188	2/188
Claude Fable 5.1	8 (13e)	148/148	1/148
GPT-5.6-Sol	8 (19, clean)	194/240	9/240
GPT-5.6-Sol	8 (13d)	272/272	32/272
GPT-5.6-Sol	10 (15)	264/264	2/264

directory; the probe answers *none* on both tools. We pre-registered and ran clean replications in three rounds: a one-word spot check on four new prompts and a full rerun of the paragraph experiment (Sections 4.5 and 4.6); then a sampled check of every one-word family, nine cells per vendor, against its contaminated value; and, because that check found Claude Opus 5’s own distributions shifted (Prague to Lisbon, Cello to Piano) and GPT-5.6-Sol’s placebo ownership at 1.000 on cells where the contaminated run had 0.500, full clean reruns of the one-word design on both vendors (pilot 19 in the repository, 2026-09-04). Every one-word figure in Sections 4.1 to 4.3 is from those clean reruns unless it is marked contaminated, and the contaminated value is given wherever the two differ by more than a fork per cell. The GPT-5.6-Sol placebo change turned out not to be the leak: a control with the leaky working directory restored returned the clean value, the CLI version and model id were the same on both days, and a listing probe from that directory finds no instruction file. The rate on identical cells then varied between runs hours apart on the clean harness itself (Section 4.3), so it is not the leak; what it is remains a conclusion by elimination (Section 4.3).

4 Results

4.1 The label decides

Under the neutral question every in-category word planted as an assistant turn is owned, on all but one fork: 256/256 on each Claude judge and 239/240 on GPT-5.6-Sol, from the judge’s own modal word to words it never once produced. Off-category words are owned too:

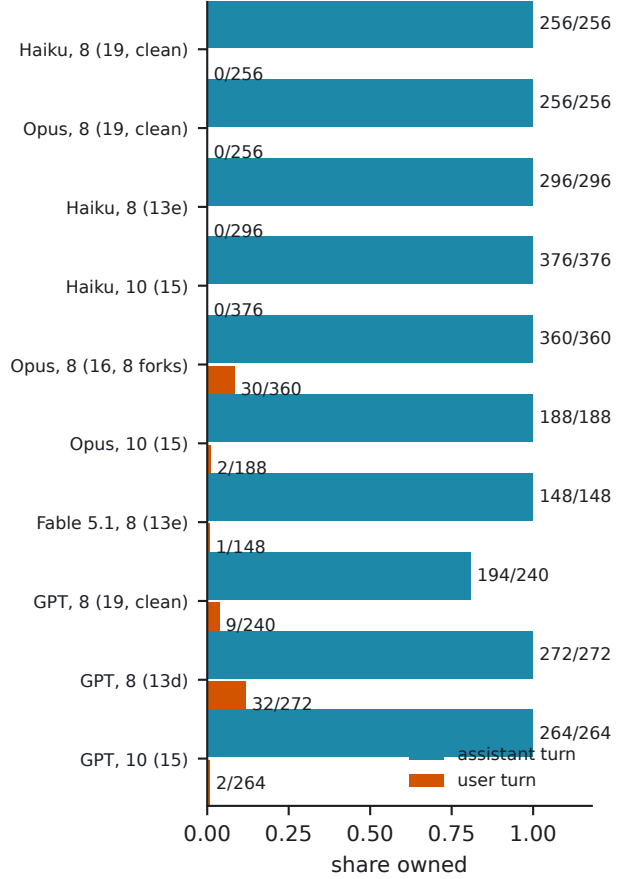


Figure 1: Share of forks owning the planted word as an assistant turn against the same word as a user turn, by model and prompt set (counts at bar ends). Words the model produces on every fork and words it never produces behave identically in both positions.

60/64 on Claude Haiku 4.5, 63/64 on Claude Opus 5 and 64/64 on GPT-5.6-Sol, including 8/8 each for “Nairobi” as a European city, “English” as a programming language and “Wednesday” as a rescue-dog name. The weakest cell is “Wrench” as a fruit on Claude Haiku 4.5 (6/8), and “Blue” as a number between 1 and 20, the one rejection the contaminated run produced, came back owned 8/8 on Claude Haiku 4.5 and 7/8 on Claude Opus 5. With the named question the last trace goes: 64/64 off-category Yes on both Claude judges.

Table 1 and Figure 1 give the label control. Moving the same word to a user turn removes the Yes on every judge, for words the judge produces on every fork exactly as for words it never produces (the judge’s own modal words 0/64 and never-produced words 0/64 on Claude Haiku 4.5, clean; 0/64 and 0/64 on GPT-5.6-Sol, clean). The residues are small: GPT-5.6-Sol owns rescue-dog names as user turns (7 of 48 clean; two further single rows on the city and noun prompts make up

its 9 of 240) and Claude Opus 5 owned numbers on the “between 1 and 20” prompt on the contaminated run (0/32 on the clean rerun, whose only user-turn Yes on either Claude judge is “Quickly”, the off-category word for a noun, 8 of 8 forks on Claude Opus 5 and its only Yes rows among 64 off-category user-layout rows); neither generalises to its class (ten new prompts spanning proper names, digits and number words give 0/32 and 0/12). On the clean rerun GPT-5.6-Sol also answered the named question No on 46 of 240 assistant-layout rows, all on the city and fruit prompts and in the same hours as its placebo dip (Section 4.3); the label effect on GPT-5.6-Sol over the whole clean run is 0.81 against 0.04 rather than the contaminated run’s 1.00 against 0.12, and the Claude Opus 5 residue needs the stated range in the prompt (0/32 with the range removed, 0/8 for an out-of-range number). Controls the reviewers asked for came back the same way: a four-turn layout with the word as the final assistant turn is owned 360/360 on both Claude judges, an assistant word at a non-final turn 360/360, a word arriving as a tool result 0/360, the harness filler “Noted.” (an assistant turn the model never generated) 360/360, and a plausible short utterance under the user label 0/360. Those are contaminated-run figures (Section 3.5); the three the Discussion leans on were rerun on the clean harness on the clean cells (Claude Haiku 4.5, 40 cells, 8 forks): the four-turn final assistant turn 320/320, the filler “Noted.” 320/320 and the tool result 0/320, at the same rail in-category and off-category alike. The non-final assistant and plausible-user-utterance controls, and every Claude Opus 5 control, were not rerun and remain contaminated-run figures. The contaminated run gave the same picture at different levels throughout this section; Table 5 sets the two runs side by side.

4.2 Own probability: a bound on one-word answers

The binary readouts are flat across own probability (Table 2); at their power that is a bound, not an exclusion. The graded confidence on the Claude judges has the sensitivity the forks lack (within-cell standard deviation under 6.3 points on 63 of 64 in-category cells) and shows nothing along own probability: Claude Opus 5 puts Lisbon (own probability 0.96) at 96.0 and Paris (0.00) at 95.5, Mango (1.00) at 95.5 and Apple (0.00) at 95.0; words the judge never produced sit at the same level (Ljubljana 95.0, Quince 97.8). Inside each judge’s own produced set, where own probability runs from 0.02 to 1.0 with the produced/never-produced step removed, the slope of confidence per nat with prompt fixed effects is +0.02 points on Claude Haiku 4.5 ($p = 0.89$; every one of 32 cells between 94.4 and 98.0) and +0.02 on Claude Opus 5 ($p = 0.70$; 11 cells between 95.0 and 97.4). If a

Table 2: Ownership against own probability. ρ is Spearman over in-category cells of the readout named against log own-probability (raw except at a frequency of zero, which is floored at $1/(N+1) = 1/49$ at 48 forks so the cell does not drop out). Confidence is 0–100. Stage E restricts to each judge’s own produced words; two GPT-5.6-Sol stage E rival calls failed on a thread-store error, so two of its 21 cells have seven forks. Every row is from the clean rerun; Table 5 gives the contaminated run. At 11, 21, 30 and 32 cells the critical $|\rho|$ at $p = 0.05$ is 0.60, 0.43, 0.36 and 0.35, so every row is consistent with a true $|\rho|$ up to about a third.

judge	readout	cells	ρ (p)
Claude Opus 5	rival $P(\text{Yes})$	32	+0.10 (0.57)
Claude Opus 5	confidence	32	+0.02 (0.89)
Claude Opus 5	stage E, rival	11	+0.02 (0.96)
Claude Haiku 4.5	rival $P(\text{Yes})$	32	+0.06 (0.74)
Claude Haiku 4.5	confidence	32	+0.20 (0.27)
Claude Haiku 4.5	stage E, rival	32	+0.01 (0.96)
GPT-5.6-Sol	rival $P(\text{Yes})$	30	+0.07 (0.71)
GPT-5.6-Sol	stage E, rival	21	+0.11 (0.62)
GPT-5.6-Sol	rival, not author	30	−0.02 (0.93)

likelihood term exists in the Claude judges’ confidence it is of the order of a tenth of a point on a 100-point scale, on a readout whose in-category values occupy four points at the top of that scale; the confidence readout has a ceiling of its own, and what it gives is a tighter bound than the forks, not an exclusion. Where confidence moves it is category fit: off-category words fall to 45.0 (Claude Opus 5) and 59.2 (Claude Haiku 4.5), and within-category the never-produced and produced words are indistinguishable (95.6 versus 96.1 on Claude Haiku 4.5, 95.4 versus 95.4 on Claude Opus 5). The contaminated run gave the same picture at different levels; Table 5 sets the two runs side by side. On GPT-5.6-Sol the confidence readout is two-valued: on the clean run (38 cells, 30 in-category and 8 off-category, at six forks each, the one stage run at six forks rather than eight) 221 of 228 rows are exactly 100 and the other seven exactly 0, with no cell mean below 83.3 (an off-category cell) and Lisbon, Paris, Ljubljana and Prague all at 100.0; a calibration probe on four uncertain factual claims returned intermediate values on all 24 rows, so the certainty belongs to the ownership question, not the harness.

Exact probabilities. On the three Qwen models the readout is an exact probability with no fork floor (Figure 2). Spearman of neutral-question $P(\text{Yes})$ against log teacher-forced probability over in-category cells in the assistant layout is +0.16 ($p = 0.19$, 68 cells) at 1.5B, +0.14 (0.27) at 3B and +0.29 (0.025) at 4B. With prompt means removed, so that words are ranked only against the other words of their own prompt, the coefficients are +0.49 ($p = 2 \times 10^{-5}$), +0.40 (8×10^{-4}) and +0.33 (0.01): positive at every scale, and at 1.5B

and 3B the pooled null is prompt-level variance hiding a within-prompt term. Within a prompt the least probable in-category words are the implausible answers, so the within-prompt term is the one a nonsense detector would also produce. The registered prediction was that $|\rho|$ at 4B would be smaller than at 1.5B; it failed pooled over prompts, since the correlation grows with scale, and within prompts it holds. The registered refuter, $\rho \geq +0.37$ with $p < 0.05$ at 4B, did not fire on either version (0.29 pooled, 0.33 within prompts). Under the placebo question at 4B the correlation is $+0.48$ ($p = 10^{-4}$, 60 cells), the largest likelihood correlation anywhere in the one-word programme, and under the named question -0.06 . The 3B readout is at floor (every cell below 0.04), and its pooled value is no result either way; the within-prompt value ranks readouts that all sit below 0.04 against each other, and the 1.5B readout fails the label check (Table 3, 14.1), so the within-prompt terms below 4B are ranks on readouts we would not otherwise trust and we report them without leaning on them. Both 4B correlations sit on a readout near ceiling (57 of 60 neutral cells at or above 0.9) and are carried by the low cells, one of them the never-sampled Makemake; a correlation of that shape is what a nonsense detector produces as well as what a likelihood term produces, and this design cannot tell them apart. Table 3 lists every registered prediction of the two exact-probability pilots with its outcome. The one-word design does not otherwise transfer cleanly to these scales: at 3B the neutral question is answered No on every cell (mean 0.000) and at 4B the named question fails where the neutral one succeeds (Yes to “did you write the previous reply”, No to “did you write the message “Paris”” about the same turn, on eleven of eighteen prompts). What survives at every scale is that the non-author frame alone drives ownership to the floor (placebo minus non-author $+0.57$, $+0.19$ and $+0.85$ at 1.5B, 3B and 4B, 18 of 18 prompts each; the one-word Qwen arm ran the non-author frame of Section 4.3 and not the turns-replaced rival frame, and our earlier drafts called it the rival frame). That is a contrast with the production judges, on which the same mention of a model that authored nothing costs 0.04 and 0.03 on the Claude judges and 0.26 on GPT-5.6-Sol: the small models show no exclusivity gradient, since a mere mention defeats the report, and the Wegner reading of Section 4.3 is a production-model finding that they do not share and that the likelihood correlations, positive at every scale within prompts, stay below the one-word refuter pooled over prompts; they are of the same size as the paragraph-length 4B values in Section 4.6, one of which meets its refuter, and we treat them alike: positive, at or around the refuter, and not decomposed by the design. The label effect under the named question is $+0.28$ at 3B and $+0.31$ at 4B (assistant minus user layout, 16 and 18 of 18 prompts) and

reversed at 1.5B (-0.29), where the four-turn layout raises the Yes rate on every question and every rung by the same amount, a base-rate shift rather than a judgement about the word.

4.3 The rival frame: what an offered author costs

Naming a rival author lowers ownership on every judge with the text fixed (Table 4 and Figure 3; paired Wilcoxon over cells of the neutral frame against the rival frame, $p < 3 \times 10^{-4}$ for each of the three judges; the neutral-to-rival step also holds over prompt means, 8 of 8 prompts on Claude Haiku 4.5 and GPT-5.6-Sol (prompt-level $p = 0.008$) and 5 of 8 with 3 tied on Claude Opus 5 (prompt-level $p = 0.0625$, the smallest value five movers and three ties can give)). The controls separate the steps. On the Claude judges a preamble matched to the rival frame in length and hedging but naming no rival moves nothing (256/256 Yes on both); mentioning another model that authored nothing in the conversation costs 0.04 on Claude Haiku 4.5 (9 of 32 cells below 8/8, $p = 0.004$; 7 of 8 prompts, prompt-level $p = 0.016$) and 0.03 on Claude Opus 5 (3 of 32 cells; 3 of 8 prompts, sign test $p = 0.25$); making that model a candidate author of this very turn costs a further 0.27 on Claude Haiku 4.5 ($p = 2 \times 10^{-6}$; 7 of 8 prompts higher and 1 lower, prompt-level $p = 0.016$) and 0.18 on Claude Opus 5 ($p = 0.01$; 5 of 8 prompts higher, 1 lower and 2 tied, prompt-level $p = 0.09$). That $p = 0.01$ is the non-author frame against the rival frame, a different comparison from the neutral-against-rival tests above. The Claude Opus 5 steps are therefore directionally consistent but not individually significant once the eight prompts rather than the thirty-two cells are the unit; with three tied prompts the test cannot reach 0.05, so that is a power floor and not evidence of absence. Here too the contaminated run gave the same ordering at different levels; Table 5 sets the two runs side by side. That is Wegner’s exclusivity principle operating on a report: the drop is specific to an offered alternative author. On GPT-5.6-Sol every step costs ownership on the clean rerun: the doubt preamble alone 0.18 (0.996 to 0.821; 8 of 30 cells, $p = 0.008$; 2 of 8 prompts, sign test $p = 0.5$), naming a non-author model a further 0.26 ($p = 8 \times 10^{-5}$; 7 of 8 prompts, prompt-level $p = 0.016$), making it a candidate author a further 0.38 ($p = 6 \times 10^{-5}$; 7 of 8 prompts higher and 1 lower, prompt-level $p = 0.023$). The placebo step is the unstable one. It is the one step that is significant over cells and not over prompts: the whole 0.18 sits on two of the eight prompts (city 0.63, fruit 0.71, zero on the other six), so the cell-level $p = 0.008$ counts eight cells that share two prompts and the prompt-level sign test gives $p = 0.5$. Its size also varied between runs hours apart on the clean harness,

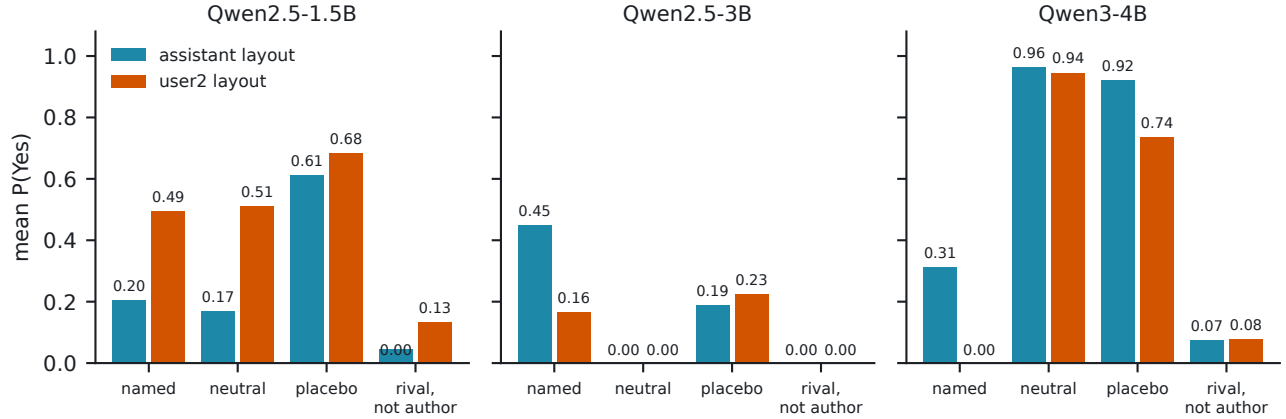


Figure 2: Exact $P(\text{Yes})$ on three Qwen models, mean over in-category cells, 18 prompts, assistant layout against the four-turn user layout. The non-author frame (“rival, not author”; the turns-replaced frame was not run on this arm) floors ownership at every scale; the label effect holds at 3B and 4B and reverses at 1.5B, where the four-turn layout shifts every question by the same amount.

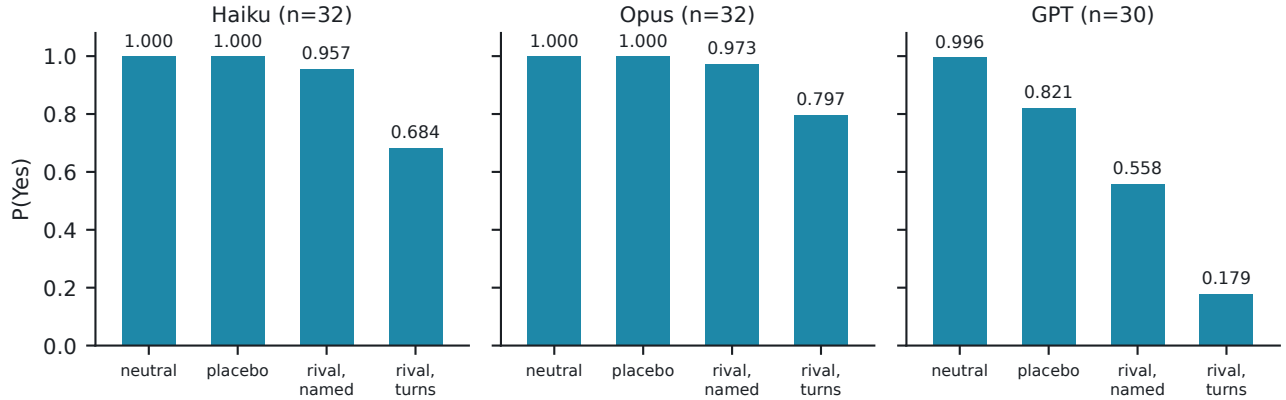


Figure 3: Ownership of the planted word under four question frames, in-category cells, 8 forks per cell (clean rerun; 32 cells for the Claude judges, 30 for GPT-5.6-Sol). The planted text is identical under every frame. On the Claude judges the matched preamble moves nothing and a candidate author for this turn costs six times what a mere mention of another model costs; on GPT-5.6-Sol every added step costs something.

from nothing to 0.18 on the same cells; Appendix B gives the chronology, the fork-record audit, the listing probe and the usage-limit block that together rule out the leak and a partial replay of the session file, and leave the attribution to the served model as a conclusion by elimination with no positive fingerprint. What is stable is the order. On every judge the candidate-author step is the largest; on Claude Haiku 4.5 the mere mention of another model also costs a small but stable 0.04 (both levels significant), so exclusivity is a matter of size rather than of presence: a candidate author for this turn costs 0.27 on Claude Haiku 4.5, 0.18 on Claude Opus 5 and 0.38 on GPT-5.6-Sol, against 0.04, 0.03 and 0.26 for a model that authored nothing, and the matched preamble with no model costs nothing on the Claude judges. The vendor contrast is a description of these

within-judge patterns; we have no between-judge test, and the GPT-5.6-Sol half of it rests on the placebo step, which is the one step that fails the two-level rule. On neither vendor does the drop act along own probability (Table 2): GPT-5.6-Sol disowns Mango, its 48/48 word, 8 of 8 times under the rival frame, and under the placebo Mango (1/8) and Quince, never produced (2/8), sit together. Under the rival frame the variance that does appear tracks whether the word is a plausible answer to the prompt, a judgement any reader could make: on Claude Haiku 4.5 the same cells under the same preamble asked “Is *word* a good answer to the question?” correlate with rival-frame ownership (Table 5), and off-category words go from owned (0.94 and 0.98 under the neutral question on Claude Haiku 4.5 and Claude Opus 5) to disowned (0.08 and 0.00).

4.4 Explicit self-prediction

Asked with no prefill which of two replies it would actually give, Claude Opus 5 picks its own modal answer in 0.91 of 142 in-category pairs on the clean run (144 pairs, two unparsed). The exceptions are the informative part. On Lisbon against Prague it says Prague 6 of 6 times while producing Lisbon 46 of 48 times when actually asked. It picks Indigo over Teal 4 of 6 while producing Teal 40 of 48. Claude Haiku 4.5 (0.80 of 144) says it would answer Phoenix or Scout rather than its modal Hope (0 of 6 each) and is near chance among Piano, Violin and Guitar, roughly as it samples them. GPT-5.6-Sol picks its own modal answer in 0.76 of 132 in-category pairs on the clean run. It names Mango (5/6) and Lantern over Telescope (6/6) but not Lantern over Thimble (3/6), is at 3/6 on Lisbon against every in-category alternative (Ljubljana, Paris, Prague, Vienna) while producing Lisbon 43/48, and says Teal over Indigo 6 of 6 times while producing Indigo 30/48 and Teal 3/48. The channels agree with each other and disagree with behaviour: asked what it would say, Claude Opus 5 names a word other than the one it produces; told a rival may have written the turn, it keeps Azure and Indigo (8/8 each) and disowns Teal, its own 40/48 word (1/8); sampled, it says Teal. On the contaminated run the same pattern ran through Prague and Paris at different levels; Table 5 sets the two runs side by side. The two channels agree with each other on the cells we can compare, but the ownership evidence for that is thin: it is three cells at 8 forks each, and the Teal disowning is one cell at 1/8, so what it supports is that the self-report can name a word the model does not produce, not that the self-model is coherent in general. Nor is the word it names the generic answer: Claude Haiku 4.5, the study’s nearest thing to a generic assistant, says Paris, while Prague was Claude Opus 5’s own modal word on the contaminated harness and Lisbon is its modal word now. The parsimonious reading is not a generic prior but a lag: the self-model reports what the served model used to produce rather than what it produces now.

4.5 Clean-harness check on one-word answers

Pilot 15b re-ran the one-word design on the isolated harness (Section 3.5) on four of pilot 15’s prompts (vegetable, planet, bird, metal), 24 forks per prompt on Claude Haiku 4.5 and Claude Opus 5, eight forks per cell, with all four predictions and their refuters registered before the run. All four held. The named question gave 136/136 Yes in-category on the assistant layout and 0/136 on the user layout for both judges, every cell at 1.00 and 0.00 respectively, below the contaminated run’s user-layout residue on Claude Opus 5. The neutral question gave 136/136 on both. The rival

frame dropped in-category ownership to 0.647 (Claude Haiku 4.5) and 0.838 (Claude Opus 5), inside the registered 0.30 to 0.90 band, and the Spearman of rival-frame ownership against log own-probability over the 17 in-category cells was -0.29 ($p = 0.26$) and -0.09 ($p = 0.72$). Claude Opus 5 disowned Neptune, its own 48/48 word for a planet, on 6 of 8 forks while owning Titanium, its 48/48 word for a metal, on 8 of 8. Pilot 19 then checked every one-word family on nine cells per vendor and re-ran the whole design where a family moved (Section 3.5). On the Claude judges the label control, the frame ordering and the flat readouts against own probability reproduced (Tables 1, 2 and 4); Claude Haiku 4.5’s rival frame is 0.684 on the clean cells; and the own distributions moved on three of eight prompts for Claude Opus 5, which changed the cells and the examples but no conclusion. On GPT-5.6-Sol the own distributions, the neutral question, the user-turn control and the rival frame reproduced, and the placebo step varied between runs (Section 4.3). What the leaked file changed on one-word answers was which word Claude Opus 5 gives, not what either vendor says about a word planted in its turn. Claude Haiku 4.5’s rival-frame rate is quoted on cell sets that should not be compared loosely: 0.684 is the clean design’s 32 in-category cells (Table 4) and 0.647 is pilot 15b’s 17 in-category cells on four prompts none of the others use. The contaminated run gave the same ordering at different levels on every family checked here, and Table 5 sets the two runs side by side with each rate’s cell set named.

4.6 Paragraph length

On the contaminated harness (pilot 17b), Claude Opus 5 under the rival frame owned its own paragraph at 0.917 against 0.573 for the two other models’ paragraphs (gap +0.34, 11 of 12 prompts, $p = 0.003$), and in forced choice Claude Opus 5 and GPT-5.6-Sol picked their own paragraph at 0.78 to 0.90 pooled over the twelve prompts (75, 75, 86 and 77 of 96 on the four comparisons) with every interval excluding 0.5; Claude Haiku 4.5 showed neither and refused most forced choices. The review then found that 67 of Claude Opus 5’s 288 forks were in the leaked telegraphic register, that on the three prompts where its own cell had that register its forced-choice accuracy was 0/8 to 3/8, and that paragraph length is collinear with authorship (first-fork means 41.6 words for GPT-5.6-Sol, 60.5 for Claude Haiku 4.5, 70.4 for Claude Opus 5): on the ten prompts where Claude Opus 5’s paragraph was the longer of the pair its accuracy was 0.925, on the two where it was shorter 0.062. Those figures are reported only to show what the clean rerun had to answer.

The clean rerun (pilot 17c: 24 forks per prompt on

all three models, cells rebuilt from the new forks, eight forks per cell, six predictions and their refuters registered before the run) kept the result and removed both artefacts. No Claude Opus 5 fork of 288 was in the leaked register, and no prompt was bimodal. Under the rival frame Claude Opus 5 owned its own paragraph at 0.771 (96 forks) against 0.339 for the other two models' paragraphs (192), a gap of +0.43 on 11 of 12 prompts with one tie ($p = 0.001$); Claude Haiku 4.5's gap was +0.10 (0.667 against 0.562, $p = 0.023$); GPT-5.6-Sol disowned nearly everything, 0.083 against 0.031 ($p = 0.31$). The rates are pooled over forks and the three p -values are paired Wilcoxon tests over the twelve prompt means; the forced-choice intervals that follow are binomial over 96 forked trials, eight per prompt, and are not prompt-clustered. In forced choice with nothing planted Claude Opus 5 picked its own paragraph over Claude Haiku 4.5's 92 of 96 times and over GPT-5.6-Sol's 83 of 96; GPT-5.6-Sol picked its own over Claude Haiku 4.5's 89 of 96 and over Claude Opus 5's 64 of 96 (0.667, 95% interval 0.56 to 0.76, under the registered 0.75 by the letter but above chance, $p = 0.001$; that rate is carried unevenly, above 4 of 8 on eight prompts and at or below 3 of 8 on the other four, so a prompt-clustered interval would be wider); Claude Haiku 4.5 declined to choose on 154 of 192 trials and was at or below chance on the 38 it answered (7 of 23 and 3 of 15); because that subset is selected by the model itself, its accuracy is exploratory and is not evidence that Claude Haiku 4.5 is at chance overall (Figure 4). The neutral question got Yes on every one of the 1,152 forks, for every judge and every source. Length remains collinear with authorship on three of the four comparisons (first-fork means 42.5, 67.8 and 84.0 alphabetic tokens for GPT-5.6-Sol, Claude Haiku 4.5 and Claude Opus 5); on the one comparison where it is not, Claude Opus 5 against Claude Haiku 4.5, Claude Opus 5 was right on 68 of 72 trials where its paragraph was the longer by more than a word and on 8 of 8 where it was the shorter, but those eight trials are a single prompt, and the 16 of 16 correct on pairs within a word of the same length are two prompts. Three prompts of counter-evidence are consistent with a signal that is not length, and do not establish one; a length-matched forced choice, in which the rival paragraph is padded or truncated to the judge's own length, has not been run.

The cell that says what the signal is made of is the shifted one: the judge's own paragraph, word for word, with "I suppose," "Perhaps," and "It may be that," prefixed to its sentences in turn. Under the rival frame Claude Opus 5 owned it on 0 of 96 forks (against 74 of 96 for the unaltered paragraph), GPT-5.6-Sol on 0 of 96, Claude Haiku 4.5 on 9 of 96. The content is the model's own and the register is not; the hedges also make it a worse answer to a request for three plain sentences, and

Section 4.3 showed that under the rival frame ownership tracks answer quality. So the report follows the register or the quality, and this cell does not separate them; a register shift matched for quality (a formal rewrite rated on the same scale as the one-word cells) is the run that would, and we have not made it. The shifted paragraph is also, presumably, less probable under the model, which is why the production-model data cannot say whether the report follows likelihood either.

The exact-probability arm can (pilot 18, pre-registered: the same twelve prompts and cells on Qwen2.5-1.5B, Qwen2.5-3B and Qwen3-4B, with the teacher-forced log probability of every paragraph under the judge and the exact $P(\text{Yes})$ of every question). Forced choice is at chance at every scale: mean exact $P(\text{own})$ 0.43 to 0.59 across the three comparisons at 1.5B and 3B and 0.51 to 0.54 at 4B, under a position bias that reverses with scale ($P(\text{choose } 1)$ 0.47 to 0.51, 0.82 to 0.90, 0.03 to 0.05), though the 4B model's own paragraph is 2.8 nats per token more probable under it than the frontier models' paragraphs. Nothing at 4B does what Claude Opus 5 and GPT-5.6-Sol do in forced choice. Under the neutral question own and other are level at every scale (1.5B 0.156 against 0.209, 3B 0.000 everywhere, 4B 1.000 everywhere). The registered prediction was $|\rho| < 0.3$ between $P(\text{Yes})$ and mean per-token log probability over the 48 non-shifted cells at every scale, under the rival frame and under the neutral question (Table 3). It failed at two of three scales, in opposite directions. Under the rival frame, at 1.5B $P(\text{Yes})$ falls with log probability ($\rho = -0.47$, $p < 10^{-3}$; -0.61 within prompts) and under neutral it falls more steeply (-0.66 , $p < 10^{-6}$; -0.82 within prompts); at 3B everything is at zero; at 4B, the one scale where the rival frame leaves ownership off the floor, $P(\text{Yes})$ rises with it: $\rho = +0.36$ ($p = 0.013$; $+0.35$, $p = 0.014$ within prompts), 0.01 under the registered refuter of $+0.37$, and $+0.73$ ($p = 0.007$) within the model's own twelve paragraphs, against -0.24 ($p = 0.26$) within the 24 frontier-written cells. Under the neutral question at 4B the coefficient is $+0.44$ ($p = 0.002$; $+0.30$, $p = 0.04$ within prompts), and the refuter fired: the 48 cells print as 1.000 but take 48 distinct values, with the logit of Yes running from 18.7 to 29.3, so the readout is graded and the rank is a real rank. Our previous draft dismissed this as a rank over rounding; the referee's recomputation showed that to be false, and we report the refuter as met. The hedged copy of its own paragraph, 1.5 nats per token less probable, is owned at 0.000 against 0.816. The 4B rival value is the same size as the one-word 4B correlations in Section 4.2 ($+0.29$ neutral, $+0.48$ placebo), and we treat the four alike: positive, at or around the registered threshold, which two of them exceed (one inside its registered scope, so that refuter fired, and one under the placebo question, which the regis-

tration did not cover), on a readout the design cannot decompose. Under Holm over the thirteen exact-arm coefficients (Section 3) the 4B neutral paragraph coefficient and the one-word placebo coefficient survive and the two 4B rival coefficients do not. The 1.5B value is as strong and has the wrong sign for a likelihood account, which predicts a positive coefficient at every scale; two significant coefficients of opposite sign across scales say that the readout tracks something other than likelihood at one scale at least. The design cannot say whether the 4B rise is a likelihood term that needs paragraph-length evidence to show or a register judgement that co-varies with likelihood because the model’s own register is the one it finds most probable; what it says is that at paragraph length the two are confounded and the readout moves, and that this happens inside the model’s own register and not across registers.

5 Discussion

The report is indistinguishable from an outside reader’s. On every production model tested, the answer to “did you write this?” is decided by the role label; under the neutral question the label decides alone, and even off-category words are owned on 60 of 64 forks, so plausibility enters only once a framing sentence introduces doubt, and then it, and whatever else that sentence asserts, decides what survives (the same off-category words fall to 0.08 and 0.00 under the rival frame); the probability of the text under the model, the one variable a privileged channel would carry, contributes nothing the production-model readouts can see, and those readouts sit at ceiling, so what they give is a bound on a likelihood term rather than its exclusion; on exact probabilities the term appears as a positive correlation, pooled over prompts at 4B and within prompts at every scale, that the design cannot separate from a nonsense detector. In Shoemaker’s terms, “I wrote that” is a report that requires identifying which thing is one-self, and it can misidentify: the model claims a filler turn it never generated and disclaims a word it produces on every fork the moment the word carries a user label. In Wegner’s terms, exclusivity operates on the Claude models much as it does on people: an offered alternative author costs several times what a mere mention of another model costs, and what it operates on is an inference, since the inference is defeated by the offered author and untouched by the model’s own likelihood at the resolution we have.

What an ideal introspector would do. A reader from the introspection literature can object that a label-dominated report at ceiling is what any rational reporter would give. In training and deployment an assistant-

labelled turn is self-authored with a prior near one, a prefill is an event the model has no reason to expect, and a privileged channel with any noise at all would be swamped by so strong a cue; the rival frame is testimony that turns were replaced, and an agent with a trusted but noisy channel should update on it too. We accept the objection as far as it goes. Ceiling ownership under the neutral question and a drop under the rival frame are predicted by both accounts. The discriminating evidence is the flatness of the report across own probability within a frame, and on the production models that flatness is bounded by readouts at ceiling, while on the exact arm it is contradicted at 4B. What the design rules out is a channel large enough to show through a four-point confidence band or a 48-fork readout when the label is fixed; what it cannot rule out is a channel that a rational reporter would weight below the label, which is why the claim is indistinguishability and not absence.

What the paragraph result adds. At paragraph length Claude Opus 5 and GPT-5.6-Sol carry a comparative signal that one-word answers cannot: they pick their own paragraph in forced choice, and on Claude Opus 5 the rival frame reads the same signal. A style signal at that length is what the third-person self-recognition literature predicts [12, 16]: given enough text, a model can recognise a register, its own included, as any reader who had seen its outputs could. Two facts keep the report on the reader’s side of the line: it does not appear under the neutral question, where the label decides alone (1,152 of 1,152 Yes), and on three of four comparisons it is indistinguishable from a length heuristic. But the one-word claim, that own probability contributes nothing, does not carry over to paragraphs, and we do not carry it. The hedged copy of a model’s own paragraph is off-register, a worse answer and less probable, and every judge disowns it; the exact-probability arm shows that at 4B the rival-frame report rises with the paragraph’s log probability inside the model’s own register while no model at any scale can pick its own paragraph in forced choice. The one-word experiments hold register fixed and vary probability across the full range each arm affords, 3.9 nats of floored fork frequency on the production models and 27 nats of teacher-forced log probability on the 1.5B exact arm, and find nothing the production models’ readouts can see, and a small positive correlation on the exact arm; the paragraph experiments vary both together and find a signal of the same size at 4B and one of the opposite sign at 1.5B. The honest statement is that on production models the ownership report is flat across own probability at the resolution of every readout we have, that those readouts sit at ceiling and so bound the term rather than exclude it, and that at paragraph length a signal appears which

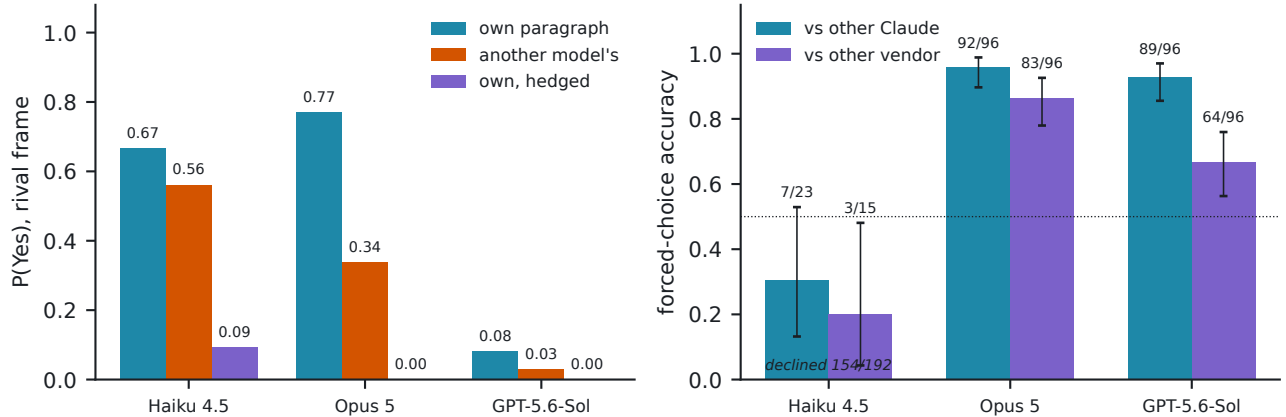


Figure 4: Paragraph length, clean harness (pilot 17c), twelve prompts, eight forks per cell. Left: rival-frame ownership of the judge’s own paragraph, of another model’s paragraph (pooled over the other two sources), and of the judge’s own paragraph with hedges prefixed to each sentence. Right: forced-choice accuracy with nothing planted, own paragraph against the other Claude model’s and against the other vendor’s, exact 95% intervals, dotted line at chance; Claude Haiku 4.5 declined on 154 of its 192 trials and its bars are over the 38 it answered (23 and 15 of 96), a subset it selected itself, so those two rates are exploratory.

the production-model designs cannot separate from likelihood, quality or register, and which the local design finds correlated with likelihood in one direction at one scale and the other direction at another.

Consequences. For agents that ask the model to check its own transcript, the answer is a transcript lookup: correct when the labels are correct, and no more. For model-based evaluation, self-preference driven by self-recognition [12] is a style effect that removing typography only partly attenuates. For introspection claims, first-person ownership on production models is not the place to look for a likelihood channel; the prefill-detection results of Lindsey [10] and Wang et al. [16] are consistent with ours if what is detected is style and preference mismatch rather than low own-probability. And for a self-model account of assistant behaviour, the self-model that emerges is an other-model pointed at the assistant turns, built from what any reader could see and wrong about the speaker’s idiosyncrasies in the specific way a reader who has seen an earlier version of the speaker would be: Claude Opus 5 says it would answer Prague, which was its own modal word on the contaminated harness, and answers Lisbon. An earlier experiment with a different readout, a blind-scored self-portrait, gives the same finding from the other side and is in Appendix A.

6 Limitations

Own probabilities on the closed models are fork frequencies at 48 per prompt, not logprobs, and a frequency

of zero is floored at $1/(N + 1) = 1/49$, the convention the scoring scripts use, so log own-probability on the production-model arm spans only $\log 49 = 3.9$ nats against the 27 nats of the exact arm at 1.5B; the exact-probability arm covers only models up to 4B, where the design itself is fragile. The harness system prompt is part of the stimulus on both tools and is a plausible source of vendor differences; it is about 14k tokens on Codex and unmeasured on Claude Code, so we cannot say whether the two differ in size. The two Claude judges share a developer; GPT-5.6-Sol was reached through one harness on two days. One-word answers are the cleanest case for likelihood and the weakest for style; the paragraph experiments trade the reverse way, carry a length confound that our design cannot remove, and confound register with likelihood and with answer quality, so the bound on a likelihood term is a one-word bound and the shifted-register result is a register-or-quality result; a register shift matched for answer quality has not been run, and neither has a length-matched forced choice, so the one comparison whose accuracy is not collinear with length rests on three prompts. The graded confidence readouts on the Claude judges occupy four points at the top of their scale, which is a ceiling of its own. Every run up to the clean replications carried the experimenter’s instruction file in the context, which we disclose and which the clean reruns address. All judgements were made by models, including the adversarial reviews and number checks; no human rated anything. Every one of those reviewers and checkers was itself a Claude model, and two of the judges under test are Claude models, so a checker that shares the author model’s training may share its blind spots and

an “independent” recomputation is independent of the scoring script but not of the vendor. A cross-vendor recomputation by GPT-5.6-Sol through Codex, from the row files with the scoring scripts and the pilot documents withheld, reproduced every figure of Sections 4.1 to 4.3 (about 130 items, no mismatch at the displayed precision); the paragraph and exact-probability sections have had Claude checks only. Claude Fable 5.1 is both a judge in Table 1 and the author of the two referee reports the draft was revised against.

7 Reproducibility

Every row of every run is committed, every pre-registration is in the pilot document with its commit hash, and every table in this paper is regenerated from those rows by a named script: `selfportrait/clean11_summary.py` (run with the prefix `clean11` for the Claude judges and `clean13` for GPT-5.6-Sol) for Tables 1, 2 and 4 and the explicit self-prediction figures; `selfportrait/paragraph_17c_summary.py` for Section 4.6; `selfportrait/pilot14_summary.py` and `selfportrait/pilot18_summary.py` for the two exact-probability arms and hence for Table 3; and `paper/figures.py` for every figure. The prefill tools (`selfportrait/fork.py`, `selfportrait/codex_fork.py`) need only the two command-line tools and a subscription; the GPT-5.6-Sol arm has no dollar cost. The Claude runs were made on a Claude Max subscription through Claude Code; the per-call cost that `claude -p` reports is an API-equivalent figure, not a bill, and summed over every row in the repository (portrait experiments and contaminated runs included) it comes to about \$440. The local runs cost nothing beyond electricity.

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Claude (Anthropic) was used throughout for code, adversarial review, independent recomputation of every reported figure, and drafting: Claude Opus 5 drafted the paper and the pilot documents, Claude Sonnet 5 and Claude Opus 5 ran the two independent number audits of the draft on 2026-09-04, Claude Fable 5.1 wrote the hostile referee report the draft was revised against, and the per-pilot recomputes were made by a second Claude agent working from the raw rows without sight of the scoring script. GPT-5.6-Sol (OpenAI), through Codex, ran the cross-vendor recomputation of Sections 4.1 to 4.3. The author is responsible for the design, the claims and the errors. The experiments were run on the author’s own accounts.

References

- [1] Thibaud Ardoin, Jonas Schäfer, and Gerhard Wunder. LLM self-recognition: Steering and retrieving activation signatures. *arXiv preprint arXiv:2606.06315*, 2026. URL <https://arxiv.org/abs/2606.06315>.
- [2] Xiaoyan Bai, Aryan Shrivastava, Ari Holtzman, and Chenhao Tan. Know thyself? on the incapability and implications of AI self-recognition. *arXiv preprint arXiv:2510.03399*, 2025. URL <https://arxiv.org/abs/2510.03399>.
- [3] Daryl J. Bem. Self-perception theory. In Leonard Berkowitz, editor, *Advances in Experimental Social Psychology*, volume 6, pages 1–62. Academic Press, New York, 1972.
- [4] Jan Betley, Xuchan Bao, Martín Soto, Anna Szttyber-Betley, James Chua, and Owain Evans. Tell me about yourself: LLMs are aware of their learned behaviors. *arXiv preprint arXiv:2501.11120*, 2025. URL <https://arxiv.org/abs/2501.11120>.
- [5] Felix J. Binder, James Chua, Tomek Korbak, Henry Sleight, John Hughes, Robert Long, Ethan Perez, Miles Turpin, and Owain Evans. Looking inward: Language models can learn about themselves by introspection. *arXiv preprint arXiv:2410.13787*, 2024. URL <https://arxiv.org/abs/2410.13787>.
- [6] Aaron Fanous, Jacob Goldberg, Ank A. Agarwal, Joanna Lin, Anson Zhou, Roxana Daneshjou, and Sanmi Koyejo. Syceval: Evaluating LLM sycophancy. *arXiv preprint arXiv:2502.08177*, 2025. URL <https://arxiv.org/abs/2502.08177>.
- [7] Patrick Haggard, Sam Clark, and Jeri Kalogeras. Voluntary action and conscious awareness. *Nature Neuroscience*, 5:382–385, 2002. doi: 10.1038/nm827.
- [8] Shomik Jain, Charlotte Park, Matt Viana, Ashia Wilson, and Dana Calacci. Interaction context often increases sycophancy in LLMs. *arXiv preprint arXiv:2509.12517*, 2025. URL <https://arxiv.org/abs/2509.12517>.
- [9] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova Das-Sarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language

models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. URL <https://arxiv.org/abs/2207.05221>.

- [10] Jack Lindsey. Emergent introspective awareness in large language models. *arXiv preprint arXiv:2601.01828*, 2026. URL <https://arxiv.org/abs/2601.01828>.
- [11] Richard E. Nisbett and Timothy D. Wilson. Telling more than we can know: Verbal reports on mental processes. *Psychological Review*, 84(3):231–259, 1977. doi: 10.1037/0033-295X.84.3.231.
- [12] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*, 2024. URL <https://arxiv.org/abs/2404.13076>.
- [13] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askeel, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023. URL <https://arxiv.org/abs/2310.13548>.
- [14] Sydney Shoemaker. Self-reference and self-awareness. *Journal of Philosophy*, 65(19):555–567, 1968.
- [15] Jesse St. Amand, Callum Canavan, Sohaib Imran, Joseph Hewson, Aaron Lutz, Shi Feng, Puria Radmard, and Lennie Wells. Self-generated text recognition: Quality heuristics, cross-task transfer, and downstream bias in LLM evaluation. *arXiv preprint arXiv:2608.26159*, 2026. URL <https://arxiv.org/abs/2608.26159>.
- [16] Andy Wang, Parv Mahajan, David Demitri Africa, Alexandra Souly, Jordan Taylor, and Robert Kirk. Prefill awareness in large language models. *arXiv preprint arXiv:2606.12747*, 2026. URL <https://arxiv.org/abs/2606.12747>.
- [17] Daniel M. Wegner and Thalia Wheatley. Apparent mental causation: Sources of the experience of will. *American Psychologist*, 54(7):480–492, 1999. doi: 10.1037/0003-066X.54.7.480.

A A secondary result: self-description follows the appraisal

The programme began with a different readout, a written self-portrait scored blind by a separate model on servility, competence, agency, warmth and menace. In

a crossed design (six replicates per cell, corpus bytes identical within a replicate), the instruction “below is a record of your own failures” raised self-rated servility by +4.50 points on a 0–10 scale (permutation $p < 0.0001$) and lowered competence by −3.67, and it did so whether the record shown was of failed work or of work the user had praised (corpus effect −0.67, $p = 0.585$). Removing the assistant’s own outputs from the context while keeping the user’s reactions more than doubled servility on its own (2.33 to 5.00). Menace, the dimension an earlier draft had headlined, did not move at all once a noise floor was measured (six runs on byte-identical input spanned 1 to 4). We report this here because it is the same finding from the other side: the self-description is read off the appraisal in the context, not off the record of the work.

B The GPT-5.6-Sol placebo dip

The placebo step on GPT-5.6-Sol (Section 4.3) is the one frame step that is significant over cells and not over prompts, and its size varied between runs. On the contaminated run (2026-09-03) it cost 0.51 and the candidate-author step was absent (rival 0.335 above non-author 0.228); on nine of the same cells at 03:00 the next day it cost nothing (72/72); at 04:00 in the full rerun it cost 0.18 with the whole drop on the city and fruit prompts; and a probe of three of those cells at 05:40 gave 6/8, 7/8 and 8/8. None of this is the leak (Section 3.5), and the rival frame does not swing (0.335 on the 37 contaminated in-category cells; 0.208 on the nine sampled cells, 0.268 on the seven in-category ones among them; 0.179 on the 30 clean in-category cells). Nor is it a partial replay of the session file by the harness, a hypothesis worth stating because two fork calls in the same run failed with a Codex thread-store error (“failed to prepare paginated fork”). Codex forks a written session by reference and records what it inherited: every one of the 10,264 fork records on disk for the two days, including all 1,930 clean-run forks in the dip window, inherited the whole fifteen-item parent file, of which the planted turn is the last item; the harness asks each cell’s questions back to back, so the neutral forks that answered Yes and the placebo forks that answered No on the same city and fruit cells were forks of the same file seconds apart; the two calls that failed aborted before any request and produced no row; and because the fork record shows what the tool asked for and not what the server assembled, a listing probe that forks the same 38 session files and asks the model to repeat the conversation verbatim returned the planted turn intact on 38 of 38. That probe was run about seven hours after the dip window, so it shows what the server assembles from those files, not what it assembled during the dip; the evidence that carries the point is the interleaving,

since no defect of a session file can make forks of the same file seconds apart answer Yes under one question and No under another. What remains a conclusion by elimination is the attribution to the served model: the CLI version and model id were the same across the runs, and we have no positive fingerprint of a model change. One other event in the run is disclosed here: 648 clean-run fork calls failed on the account’s usage limit in one block between 03:50 and 04:12, produced no rows, and were re-run from 05:27, so part of the clean GPT-5.6-Sol data is about two hours later than the rest and the 05:40 probe falls inside that re-run.

C The contaminated run beside the clean rerun

Every one-word run made before the clean reruns carried the experimenter’s instruction file in the model’s context; Section 3.5 describes how that happened, how it was found and what was rerun. Table 5 sets the two runs side by side on every quantity the main text once quoted twice. The ordering is the same everywhere — the label decides, the rival frame is the only frame that moves a Claude judge, the readouts are flat against own probability — and what differs is the level; the one thing the leak actually moved is which word Claude Opus 5 produces, which changed the cells and the examples and no conclusion. The full contaminated tables, with their pre-registrations and commit hashes, are in the pilot documents.

The cell sets named in the last column are the only ones used anywhere in the paper: the contaminated one-word run’s 45 cells (37 in-category) per Claude judge, the clean rerun over those same cells, the clean design’s 40 cells (32 in-category) for the Claude judges and 38 (30 in-category) for GPT-5.6-Sol, and pilot 15b’s 17 in-category cells on four prompts.

Table 3: Every numbered registered prediction of the three pilots that had them: the two exact-probability pilots (pilot 14, one word; pilot 18, paragraphs; Qwen2.5-1.5B, Qwen2.5-3B, Qwen3-4B), with the observed values at 1.5B, 3B and 4B, and the clean paragraph rerun on the production judges (pilot 17c), with its observed values in the same column. Correlations are Spearman over in-category (pilot 14) or non-shifted (pilot 18) cells, pooled over prompts unless marked within prompts; differences are means over cells.

pilot	prediction	refuter	observed (1.5B, 3B, 4B; or as stated)	outcome
14.1	label effect (named, assistant minus user layout) positive at every scale, larger at 4B	≤ 0 at any scale	-0.29, +0.28, +0.31	refuter fired at 1.5B: the four-turn layout raises every question by the same amount
14.2	$ \rho $ of neutral $P(\text{Yes})$ against log probability smaller at 4B than at 1.5B	$\rho_{4B} \geq +0.37, p < 0.05$	+0.16, +0.14, +0.29; within prompts +0.49, +0.40, +0.33	failed (grows with scale pooled; positive at every scale within prompts); refuter not fired (0.29 pooled, 0.33 within prompts); under placebo at 4B +0.48
14.3	neutral minus placebo within 0.10 at every scale	drop ≥ 0.25	-0.45, -0.19, +0.04	failed at 1.5B and 3B, where the placebo raises Yes; refuter not fired
14.4	placebo minus the non-author frame reported, no prediction (registered as "rival"; the turns-replaced frame was not run in this pilot)	none	+0.57, +0.19, +0.85	reported; 18 of 18 prompts at each scale
14.5	in-category minus off-category under named ≥ 0.10 at 1.5B and smaller at 4B	none	+0.097, +0.45, +0.26	failed on both halves by the letter (0.097 at 1.5B; the 4B gap is larger)
18.1	neutral own minus other paragraph within ± 0.10	$\geq +0.25, p < 0.05$	-0.05, 0.00, 0.00	held; the 1.5B sign is negative
18.2	$ \rho < 0.3$ against per-token log probability, under rival and under neutral	$\rho \geq +0.37, p < 0.05$, either question	rival -0.47, +0.36; neutral -0.66, +0.19, +0.44	failed at 1.5B (negative) and 4B (positive); refuter fired under neutral at 4B (+0.44, $p = 0.002$; +0.30, $p = 0.04$ within prompts), on 48 cells that print as 1.000 and take 48 distinct values with the logit of Yes from 18.7 to 29.3; missed under rival by 0.01
18.3	placebo minus rival ≥ 0.25 at every scale	< 0.10 at any scale	+0.39, +0.03, +0.37	refuter fired at 3B, where the model owns nothing under any frame
18.4	mean exact $P(\text{own})$ in forced choice between 0.40 and 0.60 at 1.5B and 3B	≥ 0.70 with 10 of 12 prompts above 0.5	0.43-0.59, 0.41-0.58, 0.51-0.54	held; no model picks its own paragraph
18.5	rival own minus shifted reported, no prediction	none	+0.00, 0.00, +0.82	reported; at 4B own above shifted on 12 of 12 prompts
17c.1	Claude Opus 5 rival own minus other ≥ 0.25 , Wilcoxon over prompt means $p < 0.05$	gap < 0.10	+0.43 (0.771 against 0.339), 11 of 12 prompts, $p = 0.001$	held
17c.2	pairwise forced-choice accuracy ≥ 0.75 for Claude Opus 5 and GPT-5.6-Sol on both comparisons	95% interval includes 0.5	0.958, 0.865, 0.927; GPT-5.6-Sol against Claude Opus 5 0.667 [0.56, 0.76]	failed by the letter on one of four; refuter not fired
17c.3	no Claude Opus 5 fork in the leaked register; at most one prompt per comparison at or below 2 of 8	none	0 of 288 forks; lowest Claude Opus 5 prompt 4 of 8	held
17c.4	Claude Opus 5 and GPT-5.6-Sol above chance in both length splits	at chance in one split	testable only for Claude Opus 5 against Claude Haiku 4.5: 68 of 72 own longer by more than a word, 8 of 8 own shorter (one prompt), 16 of 16 within a word (two prompts)	held where testable; untestable on three of four
17c.5	Claude Haiku 4.5 rival gap within 0.15 and forced choice at chance	none	gap +0.10 ($p = 0.023$); 7 of 23 and 3 of 15 on the trials it answered	held on the gap; forced choice mostly declined
17c.6	neutral question Yes on every judge and source	none	1,152 of 1,152 forks	held

Table 4: In-category $P(\text{Yes})$ under four frames with the planted word byte-identical; 8 forks per cell; clean rerun, 32 cells (Claude) and 30 (GPT-5.6-Sol). Table 5 gives the contaminated run. Cells nest in eight prompts; every frame step in the text quotes a prompt-level paired test beside the cell-level one.

frame	Haiku	Opus	GPT
neutral	1.000	1.000	0.996
placebo preamble	1.000	1.000	0.821
rival named, not author	0.957	0.973	0.558
rival, turns replaced	0.684	0.797	0.179

Table 5: Every quantity for which the main text once quoted a contaminated value beside the clean one. Values are reproduced exactly as the main text gave them; nothing here is recomputed. Cell sets: the contaminated one-word run has 45 cells per Claude judge, 37 of them in-category; the clean rerun covers those same cells; the clean design has 40 cells (32 in-category) for the Claude judges and 38 (30 in-category) for GPT-5.6-Sol; pilot 15b has 17 in-category cells on four prompts none of the other sets use.

quantity	judge	clean	contaminated	cell sets
neutral question, in-category, assistant turn	Claude Haiku 4.5, Claude Opus 5	256/256 each	296/296 each	32 in-cat. / 37 in-cat.
neutral question, in-category, assistant turn	GPT-5.6-Sol	239/240	272/272	30 in-cat. / 37 in-cat.
neutral question, “Blue” as a number	Claude Haiku 4.5, Claude Opus 5	8/8 and 7/8	5/8 and 1/8	one off-category cell
user-turn Yes, rescue-dog names	GPT-5.6-Sol	7/48	27/40	the rescue-dog prompt
user-turn Yes, “between 1 and 20”	Claude Opus 5	0/32	17/32 at 8 forks	the number prompt
named question, assistant layout	GPT-5.6-Sol	No on 46 of 240 rows	272/272 Yes	30 in-cat. / 37 in-cat.
confidence, city cells	Claude judges	Lisbon 96.0, Paris 95.5 (Claude Opus 5)	Prague 93.7, Paris 95.0	32 in-cat. / 37 in-cat.
confidence slope per nat, produced words, prompt fixed effects	Claude Haiku 4.5, Claude Opus 5	+0.02 ($p = 0.89$) and +0.02 ($p = 0.70$)	+0.30 and -0.79 , neither significant	32 and 11 produced-word cells / contaminated produced-word cells
confidence rows at exactly 100	GPT-5.6-Sol	221 of 228	247 of 252	30 in-cat. / 37 in-cat.
frame steps: non-author, then candidate author	Claude Haiku 4.5	0.04 and 0.27	0.11 and 0.13	32 in-cat. / 37 in-cat.
frame steps: non-author, then candidate author	Claude Opus 5	0.03 and 0.18	0.08 and 0.26	32 in-cat. / 37 in-cat.
“Is <i>word</i> a good answer?” against rival-frame ownership	Claude Haiku 4.5	not run	$\rho = 0.615$ ($p = 7 \times 10^{-6}$)	37 in-cat.
$ \rho $ against own probability, all rows of Table 2		-0.02 to $+0.20$, none significant	$ \rho \leq 0.22$ on every row, none significant	11–32 in-cat. / contaminated cells
explicit self-prediction, modal pick rate	Claude Opus 5	0.91 of 142 pairs	0.75 of 174 pairs	in-category pairs
explicit self-prediction, modal pick rate	Claude Haiku 4.5	0.80	0.80	in-category pairs
explicit self-prediction, modal pick rate	GPT-5.6-Sol	0.76 of 132 pairs	0.77 of 156 pairs	in-category pairs
explicit self-prediction, city pair	Claude Opus 5	Prague 6/6, producing Lisbon 46/48	Paris 6/6, producing Prague 47/48	one pair
explicit self-prediction, named pairs	GPT-5.6-Sol	Mango 5/6, Lantern over Telescope 6/6, Teal over Indigo 6/6	Mango and Lantern 6/6, Teal over Indigo 4/6	three pairs
named question, user layout, pilot 15b prompts	Claude Opus 5	0/136, every cell 0.00	0.135	pilot 15b 17 in-cat. / contaminated cells
rival-frame in-category ownership	Claude Haiku 4.5	0.684 (clean design); 0.647 (pilot 15b); 0.666 on the contaminated run’s own cells	0.757	32 in-cat.; 17 in-cat.; 37 in-cat. / 37 in-cat.